

Question Bank

Unit – 2

1. Explain cloud enable technologies with a define examples

Answer:

Cloud Enable Technologies are the technologies, tools, and techniques that make cloud computing possible. They help organizations build, deploy, manage, and deliver applications and services over the internet instead of local computers or servers.

In simple words:

Cloud enable technologies are the building blocks that allow cloud computing to work efficiently. These technologies provide scalability, virtualization, automation, connectivity, and storage for cloud services.

Types of Cloud Enable Technologies

1. Broadband Networks and Internet Architecture

Definition

Broadband networks provide high-speed internet connectivity that allows users to access cloud services from anywhere.

Cloud computing depends on fast and reliable internet communication between users and remote cloud servers.

Key Features

- High data transfer speed
- Continuous connectivity
- Large bandwidth
- Supports real-time services

Examples

- Fiber optic internet
- 4G / 5G networks
- Wi-Fi broadband

Real Life Example

When you store files in Google Drive or watch movies on Netflix, your device communicates with cloud servers through broadband internet.

2. Data Center Technology

A Data Center is a physical facility that contains servers, storage systems, networking equipment, and power systems used to host cloud services.

Cloud providers maintain large data centers where all user data and applications are stored.

Components

- Servers
- Storage devices
- Cooling systems
- Power supply
- Network infrastructure

Examples

Major companies operating data centers:

- Amazon
- Microsoft
- Google

Real Life Example

When you upload a photo to Google Photos, the image is stored in Google's data centers located around the world.

3. Virtualization Technology

Virtualization is the technology that allows multiple virtual machines (VMs) to run on a single physical computer.

It divides one physical server into several virtual servers, each acting like an independent system.

Advantages

- Better resource utilization
- Cost reduction
- Easy scalability
- Isolation between applications

Types of Virtualization

- Server virtualization
- Storage virtualization

- Network virtualization
- Desktop virtualization

Examples

- VMware
- VirtualBox
- Hyper-V

Real Life Example

A single physical server in a cloud data center can run 10–50 virtual machines, each hosting different websites or applications.

4. Web Technology

Web technologies enable communication between users and cloud applications through web browsers.

These technologies allow cloud applications to be accessed without installing software on local devices.

Key Technologies

- HTML
- CSS
- JavaScript
- REST APIs
- HTTP / HTTPS protocols

Examples

- Gmail
- Google Docs
- Dropbox web interface

Real Life Example

You can open Google Docs in Chrome browser and edit documents online without installing Microsoft Word.

5. Multitenant Technology

Multitenancy is a cloud architecture where multiple users (tenants) share the same cloud resources but their data remains isolated and secure.

Key Characteristics

- Shared infrastructure
- Separate data for each user
- Efficient resource utilization

Advantages

- Reduced cost
- High efficiency
- Easy maintenance

Example

- Gmail servers store emails of millions of users on shared infrastructure, but each user's data is private.

2. What is a broadband network and discuss the architecture of refining third party internet service providers

A broadband network is defined as a high-speed communication infrastructure that provides continuous, high-capacity internet access. Unlike traditional narrowband or dial-up systems, broadband utilizes a wide range of frequencies, enabling the simultaneous transmission of multiple data signals including voice, video, and data. Key characteristics include an "always-on" connection that eliminates the need for dialing, and high transmission speeds ranging from Mbps to Gbps. These networks are supported by various technologies such as **Digital Subscriber Line (DSL)**, **Cable**, **Fiber Optics (FTTH)**, and wireless solutions like **4G/5G** and **Wi-Fi**.

Architecture for Delivering Third-Party ISP Services:

The architecture for delivering third-party Internet Service Provider (ISP) services describes a collaborative model where a primary network infrastructure provider owns the physical assets while third-party ISPs lease this infrastructure to serve end-users. This framework is comprised of several critical components: the **Access Network**, which connects the user to the provider via fiber or DSL; the **Aggregation Network**, which uses routers and switches to collect and forward traffic; and the **Core Network**, acting as the high-speed backbone. The **Point of Presence (PoP)** serves as the vital gateway where third-party ISPs interface with the infrastructure to provide their specific services.

Classification of ISP Architectures:

Third-party ISP delivery can be categorized into three primary models based on the level of control provided. In **Layer 2 (L2) Bitstream Access**, the provider offers access at the data link layer, allowing the ISP to manage authentication and IP addressing. Conversely,

Layer 3 (L3) Access involves the infrastructure provider managing the routing, which is easier to implement but offers less flexibility for the ISP. Finally, the **Virtual ISP Model** allows providers to operate entirely on leased infrastructure, focusing exclusively on branding, pricing, and customer service.

Operational Workflow and Strategic Impact:

The working process begins when a user connects via broadband; the resulting traffic travels through the access and aggregation networks until it reaches the PoP. At this stage, the traffic is handed over to the chosen ISP for user authentication and internet access. This architectural approach offers significant advantages, such as encouraging market competition, reducing redundant infrastructure costs, and improving service quality for consumers. However, it also introduces challenges, including increased network management complexity, heightened security concerns between multiple providers, and a heavy dependency on the primary infrastructure owner to maintain **Quality of Service (QoS)**.

3. Is connectivity a major challenge for cloud enabled services. Discuss the types of connectivity with an architecture diagram and a live case study.

Connectivity in cloud computing refers to the network mechanisms that enable communication between users, devices, on-premises infrastructure, and cloud service providers for accessing cloud resources and services. Connectivity is a major challenge in cloud-enabled services because cloud systems rely entirely on network access for operation. The reasons include:

- · Dependence on continuous network availability
- · Performance affected by latency and bandwidth
- · Network failures lead to service interruption
- · Data transmission over networks introduces security risks
- · Limited or unstable connectivity in remote locations

Types of Connectivity in Cloud Computing

1. Public Internet Connectivity

- · Cloud services are accessed through the public Internet
- · Commonly used for web applications and SaaS services

Advantages:

- · Low cost
- · Easy implementation

Disadvantages:

- · Higher latency
- · Lower security compared to private connections

2. Virtual Private Network (VPN)

- · Secure connection established over the public Internet
- · Data is encrypted during transmission

Advantages:

- · Improved security
- · Cost-effective compared to dedicated links

Disadvantages:

- · Dependent on Internet performance
- · Additional latency due to encryption

3. Dedicated Connectivity

- · Private connection between user infrastructure and cloud provider
- · Examples include AWS Direct Connect and Azure ExpressRoute

Advantages:

- · Low latency
- · High reliability
- · Enhanced security

Disadvantages:

- · High cost
- · Complex setup

4. Hybrid Connectivity

- · Combination of on-premises systems and cloud infrastructure
- · Uses VPN or dedicated connections

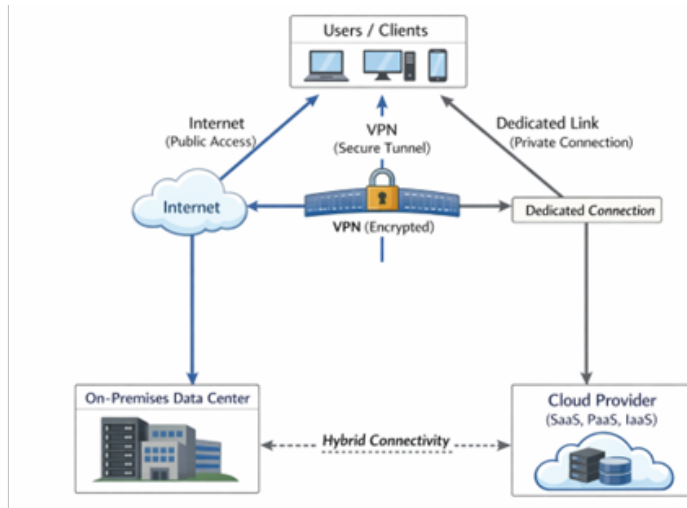
Advantages:

- · Flexibility in deployment
- · Better control over sensitive data

Disadvantages:

- Complex management
- Requires careful configuration

Architecture Diagram:



Case Study: Netflix

Netflix uses cloud infrastructure to deliver video streaming services to users worldwide.

Connectivity Usage

- Users access content through the Internet
- Content is delivered through a Content Delivery Network (CDN) called Open Connect
- Data is served from the nearest server to reduce delay

Challenges

- High bandwidth requirements for video streaming
- Network congestion during peak usage

Solutions

- Deployment of distributed servers globally
- Adaptive streaming based on available bandwidth

Outcome

- Reduced buffering
- Improved user experience through efficient content delivery

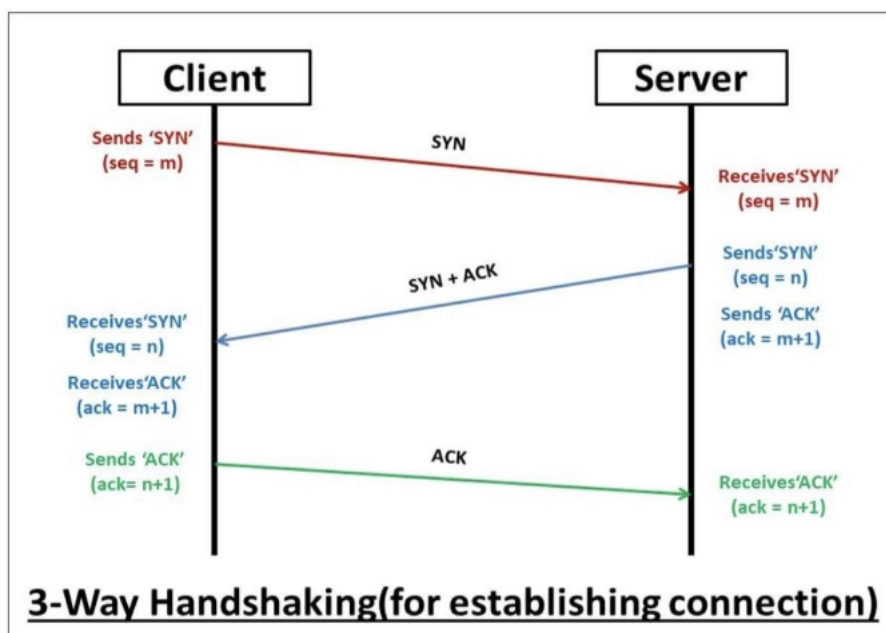
Conclusion

Connectivity is a key challenge in cloud computing as it directly impacts performance, availability, and security. Proper use of connectivity types ensures efficient cloud service delivery.

4. Is TCP a standalone protocol. Discuss the benefits and role of TCP/UDP in building sustainable cloud environment.

TCP (Transmission Control Protocol)

TCP is not a standalone protocol, as it works as part of the TCP/IP protocol suite and relies on IP for actual data delivery. It is a connection-oriented protocol that first establishes a connection using a three-way handshake (SYN, SYN-ACK, ACK). Once the connection is established, data is transmitted in the form of segments with sequence numbers, and the receiver sends acknowledgments for each segment. TCP ensures reliability through retransmission of lost packets, flow control using a sliding window mechanism, and congestion control to manage network traffic efficiently. After successful data transfer, the connection is terminated using a FIN-ACK sequence. In cloud environments, TCP plays a vital role in applications such as web services, databases, and APIs where reliable communication is required. Its benefits for a sustainable cloud include reduced data loss, efficient bandwidth usage, and minimized resource wastage due to controlled and accurate data transmission.



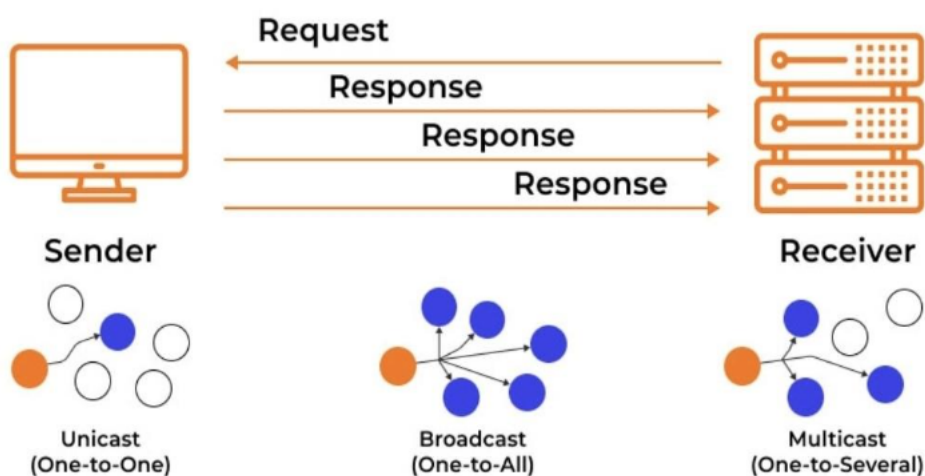
UDP (User Datagram Protocol)

UDP is a connectionless protocol that does not establish a connection before sending data. It directly transmits data in the form of datagrams without acknowledgments, sequencing,

or retransmission mechanisms. As a result, packets may be lost or arrive out of order, but this reduces overhead and increases speed. UDP is widely used in cloud environments for real-time applications such as video streaming, online gaming, DNS, and IoT systems where low latency is critical. Its benefits in building a sustainable cloud environment include lower bandwidth consumption, faster data transmission, reduced processing overhead, and improved energy efficiency, making it ideal for scalable and high-performance systems.



How User Datagram Protocol (UDP) Works



5. With an architecture explain the purpose of private cloud and public cloud.

1. Introduction to Cloud Computing

Cloud computing is a model that enables users to access computing resources such as servers, storage, databases, networking, and software over the internet instead of owning physical infrastructure.

There are mainly two important deployment models:

- Private Cloud
- Public Cloud

Each has its own architecture, purpose, advantages, and use cases.

2. Private Cloud

2.1 Definition

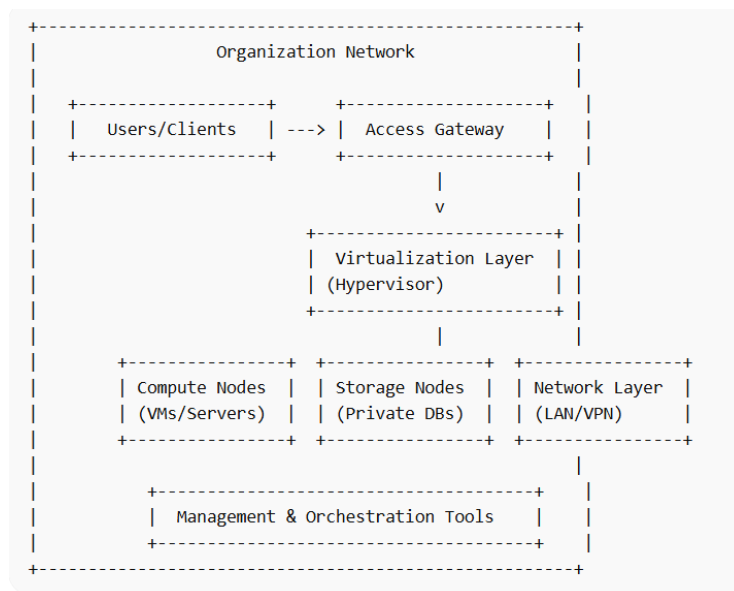
A Private Cloud is a cloud computing environment that is dedicated to a single organization. It can be hosted:

- On-premises (within the company's data center)

- Or by a third-party provider (but still exclusive to one organization)

2.2 Architecture of Private Cloud

Basic Architecture Components



2.3 Explanation of Architecture

1. Users/Clients
 - Employees access services via internal systems or VPN.
2. Access Gateway
 - Controls authentication and secure access.
3. Virtualization Layer
 - Uses hypervisors (like VMware, Hyper-V) to create virtual machines.
4. Compute Nodes
 - Physical servers running applications and workloads.
5. Storage
 - Private databases and storage systems (NAS/SAN).
6. Networking
 - Internal secure networks (LAN, VPN).
7. Management Tools

- Monitor, allocate, and manage resources.

2.4 Purpose of Private Cloud

The primary purpose of a private cloud is:

1. Enhanced Security

- Data is not shared with others
- Ideal for sensitive data (banking, healthcare)

2. Full Control

- Organization controls infrastructure, policies, and security

3. Customization

- Tailored to specific business needs

4. Compliance

- Helps meet legal and regulatory requirements

5. Performance Stability

- No resource sharing → consistent performance

2.5 Advantages of Private Cloud

- High security and privacy
- Better performance control
- Custom configurations
- Compliance-friendly

2.6 Disadvantages

- High cost (setup + maintenance)
- Requires skilled IT team
- Limited scalability compared to public cloud

2.7 Use Cases

- Banking systems
- Government organizations
- Healthcare systems
- Large enterprises with sensitive data

3. Public Cloud

3.1 Definition

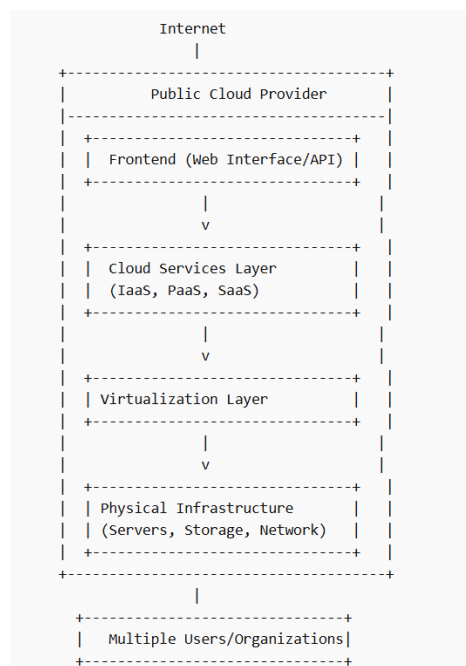
A Public Cloud is a cloud computing model where services are provided over the internet by third-party providers and shared among multiple users (multi-tenant).

Examples:

- AWS
- Microsoft Azure
- Google Cloud

3.2 Architecture of Public Cloud

Basic Architecture Diagram



3.3 Explanation of Architecture

1. Frontend

- User interface (web portals, APIs)
- 2. Cloud Services Layer
 - IaaS: Infrastructure (VMs, storage)
 - PaaS: Platforms for development
 - SaaS: Software applications
- 3. Virtualization Layer
 - Enables resource sharing across multiple users
- 4. Physical Infrastructure
 - Data centers owned by cloud provider
- 5. Multi-tenancy
 - Resources are shared among users

3.4 Purpose of Public Cloud

The public cloud is designed to:

1. Provide Cost-Effective Computing
 - No need to buy hardware
 - Pay-as-you-go model
2. Enable Scalability
 - Easily scale resources up/down
3. Provide Global Access
 - Access services from anywhere
4. Support Innovation
 - Quick deployment of applications
5. Reduce IT Management
 - No need to maintain infrastructure

3.5 Advantages

- Low initial cost

- High scalability
- Easy access
- Automatic updates and maintenance

3.6 Disadvantages

- Less control over data
- Security concerns (shared environment)
- Dependency on internet
- Possible downtime

3.7 Use Cases

- Startups and small businesses
- Web applications
- Mobile apps
- Data analytics
- E-commerce platforms

4. Key Differences Between Private and Public Cloud

Feature	Private Cloud	Public Cloud
Ownership	Single organization	Third-party provider
Security	High	Moderate
Cost	High	Low (pay-as-you-go)
Scalability	Limited	Highly scalable
Control	Full control	Limited control
Accessibility	Restricted	Global access
Customization	High	Limited

Both Private Cloud and Public Cloud play important roles in modern computing.

- Private Cloud is best for organizations that need security, control, and compliance.
- Public Cloud is ideal for businesses looking for cost-efficiency, scalability, and flexibility.

In real-world scenarios, many companies use a Hybrid Cloud, combining both models to get the best of both worlds.

7. Is bandwidth a major concern for cloud based connectivity. Justify with caase study.

Bandwidth in Cloud Connectivity

Bandwidth refers to the amount of data that can be transmitted over a network in a given amount of time (usually measured in Mbps or Gbps). In cloud computing, all services (storage, applications, processing) depend on internet connectivity, so bandwidth directly affects performance.

Is Bandwidth a Major Concern? – Yes, Justification

Bandwidth becomes a critical factor in cloud environments due to the following reasons:

1. Performance and Speed

- Low bandwidth leads to slow data transfer between users and cloud servers.
- Applications like video streaming, cloud gaming, and SaaS platforms require high bandwidth.

2. Latency and Delay

- Insufficient bandwidth increases latency, causing delays in communication.
- Real-time applications (e.g., video conferencing, online collaboration) are highly affected.

3. Data-Intensive Applications

- Big data analytics, AI/ML models, and backups require transferring large datasets.
- Limited bandwidth can bottleneck the entire system.

4. Scalability Issues

- As users increase, bandwidth demand grows.
- Without proper bandwidth, cloud systems cannot scale efficiently.

5. Cost Factor

- Higher bandwidth usage may increase operational costs for organizations using cloud services.

Case Study: Netflix (Cloud-Based Streaming Service)

Overview

Netflix is a cloud-based video streaming platform that delivers content globally using cloud infrastructure (primarily AWS).

Bandwidth Challenges Faced

- Streaming HD/4K videos requires high bandwidth (5–25 Mbps per user).
- Millions of users streaming simultaneously creates massive bandwidth demand.
- Poor bandwidth leads to:
 - o Buffering
 - o Reduced video quality
 - o Bad user experience

Solutions Implemented by Netflix

1. Content Delivery Network (CDN)

- o Netflix uses its own CDN called Open Connect.
- o Stores content closer to users to reduce bandwidth usage over long distances.

2. Adaptive Bitrate Streaming

- o Automatically adjusts video quality based on user's available bandwidth.
- o Ensures smooth playback even on slower connections.

3. Data Compression

- o Uses advanced compression techniques to reduce data size without losing quality.

Outcome

- Efficient bandwidth utilization
- Reduced buffering and latency
- Improved user experience globally

Bandwidth is a major concern in cloud-based connectivity because it directly impacts performance, scalability, and user experience. The Netflix case study clearly shows that without proper bandwidth management, cloud services cannot function efficiently. Therefore, optimizing bandwidth through techniques like CDNs and adaptive streaming is essential for successful cloud implementation.

8. What is virtualisation. Give the benefits and drawbacks of virtualisation.

Definition

Virtualization is the technique of creating a virtual version of computing resources such as servers, storage devices, networks, or operating systems using software. It allows multiple virtual machines to run on a single physical machine through a software layer called a hypervisor.

Explanation

In virtualization, the physical hardware is abstracted and divided into multiple virtual environments. Each virtual machine operates as an independent system with its own operating system and applications. The hypervisor manages resource allocation such as CPU, memory, and storage among the virtual machines.

There are two types of hypervisors:

- Type 1 (Bare-metal hypervisor): Runs directly on the physical hardware
- Type 2 (Hosted hypervisor): Runs on top of an existing operating system

Benefits of Virtualization

- Efficient Resource Utilization

Multiple virtual machines share the same physical hardware, reducing wastage.

- Cost Reduction

Decreases the need for physical servers, lowering hardware and maintenance costs.

- Isolation

Each virtual machine operates independently, preventing interference.

Virtual machines can be created, modified, or removed easily.

- Improved Disaster Recovery

Virtual machines can be backed up and restored quickly.

- Support for Testing and Development

Multiple operating systems can run on a single system for testing purposes.

Drawbacks of Virtualization

- Performance Overhead

The hypervisor introduces additional processing, which may reduce performance.

- Single Point of Failure

Failure of the host system affects all virtual machines.

- Complex Management

Requires skilled administrators to manage virtual environments.

- Security Issues

Vulnerabilities in the hypervisor can impact multiple virtual machines.

- High Initial Setup Cost

Requires investment in virtualization tools and infrastructure.

- Resource Contention

Multiple virtual machines sharing resources may lead to reduced performance

9. Explain cloud storage. Justify SAN and NAS with an example

Cloud storage is a model of storing data on remote servers that are accessed over the internet instead of local storage devices (like hard drives). These servers are managed by cloud service providers such as Amazon Web Services, Google Cloud, and Microsoft Azure.

Key Features:

- **Scalability:** Storage can be increased or decreased as needed
- **Accessibility:** Data can be accessed anytime, anywhere via the internet
- **Cost-effective:** Pay only for the storage you use
- **Data Backup & Recovery:** Automatic backups and disaster recovery options
- **Security:** Encryption and access control mechanisms

Example:

Storing files in Google Drive or Dropbox instead of saving them on your computer.

1. SAN (Storage Area Network)

Definition:

SAN is a high-speed network that connects multiple storage devices to servers, providing block-level storage.

Characteristics:

- Provides **block-level access**
- High performance and low latency
- Used in enterprise environments
- Dedicated network for storage

Example:

A large bank uses SAN to store customer transaction data. Multiple servers access the same storage system for fast processing.

Justification (Why SAN?):

- Ideal for applications requiring **high speed and performance** (e.g., databases)
- Efficient for handling large volumes of data
- Supports centralized storage management

2. **NAS (Network Attached Storage)**

Definition:

NAS is a storage device connected to a network that provides file-level storage to multiple users.

Characteristics:

- Provides **file-level access**
- Easy to set up and manage
- Works over standard network (LAN)
- Cost-effective compared to SAN

Example:

A company uses NAS to store shared files so employees can access documents from a central location.

Justification (Why NAS?):

- Best for **file sharing and collaboration**
- Simple and user-friendly
- Suitable for small to medium businesses

Difference Between SAN and NAS

Feature	SAN	NAS
Storage Type	Block-level	File-level
Speed	High	Moderate
Cost	Expensive	Affordable
Usage	Databases, enterprise apps	File sharing, backups
Network Type	Dedicated network	Standard LAN

10. What is the purpose of virtualisation technology. Describe the fundamental entities associated with the technology.

Purpose of Virtualization Technology

Virtualization technology is used to create virtual versions of physical resources such as servers, storage, networks and operating systems.

Main Purposes

- Efficient resource utilization – run multiple systems on one hardware
- Cost reduction – less hardware and power consumption
- Scalability – easy to increase or decrease resources
- Isolation & security – failure in one VM does not affect others
- Flexibility – supports different OS and applications
- High availability – supports backup, recovery and live migration

Overall purpose: To improve performance, reduce cost and provide flexible cloud infrastructure.

Fundamental Entities of Virtualization

1. Host Machine (Physical System)

The real hardware such as CPU, RAM, storage and network

Provides resources for virtualization

2. Hypervisor (Virtual Machine Monitor)

Software layer that creates and manages virtual machines

Controls resource allocation

3. Virtual Machine (VM)

Software-based computer running guest operating system and applications

Works like an independent system

4. Guest Operating System

OS installed inside VM (Windows, Linux etc.)

5. Virtual Resources

Virtual CPU, virtual memory, virtual disk, virtual network Allocated by hypervisor

11. What is virtualisation management. With a eat architecture explain the role of hypervisor in building standalone cloud system or environment

Virtualization management is the process of creating, controlling, and monitoring virtual resources such as virtual machines, storage, and networks using software tools.

It improves hardware utilization by allowing multiple virtual systems to run on a single physical machine. Key functions include:

- · Virtual machine creation and deletion
- · Resource allocation (CPU, memory, storage)
- · Performance monitoring

Architecture of Virtualization

The virtualization architecture consists of the following layers:

1. Physical Hardware Includes CPU, memory, storage, and network devices.
2. Hypervisor (VMM) Software layer that manages virtual machines.
3. Virtual Machines Each VM contains a guest operating system and applications.
4. Applications User-level programs running inside VMs.

Textual Diagram Description

- · Bottom: Physical Hardware
- · Above: Hypervisor
- · Above: Multiple VMs (each with Guest OS + Apps)
- · Top: Applications

Role of Hypervisor in Standalone Cloud

The hypervisor is the core component in a standalone cloud system.

Functions:

- · Allocates hardware resources to multiple VMs
- · Provides isolation between VMs
- · Acts as an interface between hardware and VMs
- · Enables creation and management of VMs

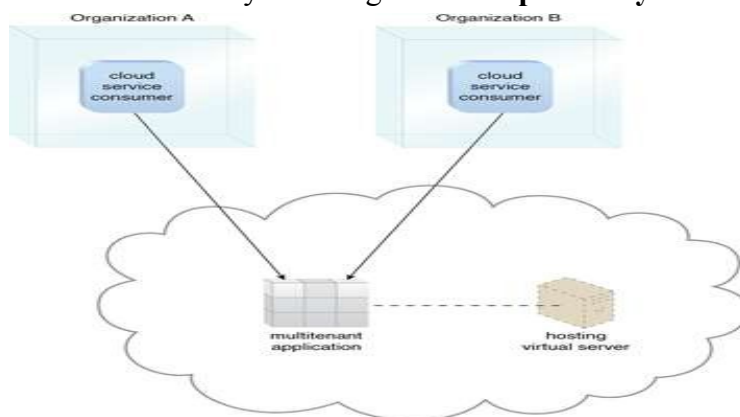
In Cloud Environment:

- · Supports resource sharing
- · Enables on-demand VM creation
- · Ensures efficient use of hardware

12. What is multi tenant technology. With a neat diagram Justify with a supporting example

Multi-tenant technology is a software architecture in which a **single instance of an application serves multiple users (tenants)**. Each tenant has its own data, but the application and infrastructure are shared.

- **Tenant** = a user/group/organization using the system
- Each tenant feels like they are using their **own private system**



Key Features

- **Shared resources** (CPU, memory, application)
- **Data isolation** between tenants
- **Cost-effective** (no need for separate servers)
- **Scalable** (easy to add new users)
- **Centralized maintenance** (updates applied once)

Supporting Example

A popular example is **Salesforce**.

- Multiple companies use Salesforce CRM.
- All customers share the same application.
- Each company's data is stored separately and securely.
- Updates are applied globally without affecting individual users.

Justification (Why Multi-Tenant is Useful)

1. Cost Efficiency

Companies don't need to maintain separate infrastructure.

2. Easy Maintenance

Software updates are done once for all users.

3. **Better Resource Utilization**

System resources are shared efficiently.

4. **Scalability**

New tenants can be added without major changes.

Advantages:

1. **Cost Efficient**
2. **Easy Updates**
3. **High Scalability**
4. **Maintenance Friendly**

Disadvantages

1. **Security Risks**
2. **Performance Issues**
3. **Limited Customization**

13. What is REST. Provide detailed validation of REST protocols with an example.

REST (Representational State Transfer) is a style used to design web services that communicate over the internet. It was introduced by Roy Fielding. In REST, data and functionality are treated as resources, and each resource is identified by a unique URL. Communication between client and server is mainly done using HTTP methods like GET, POST, PUT, and DELETE.

Basic Principles of REST

- **Client–Server:** The client (user interface) and server (data storage) are separate, which makes the system easier to manage.
- **Stateless:** Each request is independent. The server does not store any client information between requests.
- **Cacheable:** Responses can be stored and reused to improve performance.
- **Uniform Interface:** Standard HTTP methods are used to interact with resources.

Validation in REST

Validation ensures that requests sent by the client are correct and can be properly processed by the server.

- URL (URI) Validation The URL should correctly represent the resource. Example: /users/10 refers to a specific user.
- HTTP Method Validation Correct method should be used:
 - GET → fetch data
 - POST → create data
 - PUT → update data
 - DELETE → remove data
- Input Data Validation The data sent must be in proper format and valid. Example:

```
{ "name": "Ajay", "age": 20 }
```

Here, name should not be empty and age should be positive.

- Header Validation Headers like Content Type: application/json ensure correct data format.
- Status Code Validation Server must return proper responses:
 - 200 → Success
 - 201 → Created
 - 400 → Bad Request
 - 404 → Not Found

Example

Consider a Student API:

- Create Student POST /students If data is valid → 201 Created If invalid → 400 Bad Request
- Get Student GET /students/1 If exists → 200 OK If not → 404 Not Found

REST is simple and widely used for building web services. Proper validation of URLs, methods, data, headers, and responses ensures the system works correctly and avoids errors.

14. What are agents. Give/list various agents associated in the cloud operations.

Agents are autonomous software entities that perceive their environment, make decisions, and take actions to achieve specific goals. In cloud operations, agents help automate monitoring, management, security, and optimization tasks.

- Agents in Cloud Operations

Here are the main types:

1. Monitoring Agents

- Collect metrics (CPU, memory, network usage)
- Example: AWS CloudWatch Agent

2. Logging Agents

- Gather and forward logs for analysis
- Example: Fluentd, Logstash

3. Security Agents

- Detect threats, vulnerabilities, and unauthorized access
- Example: Endpoint detection & response (EDR) agents

4. Configuration Management Agents

- Maintain system configurations automatically
- Example: Ansible agents, Puppet agents

5. Deployment Agents

- Handle application deployment and updates
- Example: CI/CD pipeline agents

6. Backup Agents

- Manage automated backups and recovery
- Example: Cloud backup services agents

7. Scaling (Auto-scaling) Agents

- Automatically adjust resources based on demand

8. Cost Optimization Agents

- Monitor usage and suggest cost-saving measures

9. Networking Agents

- Manage routing, load balancing, and network policies

10. AI/Intelligent Agents (AIOps)

Use AI to predict failures, optimize performance, and automate decisions

15. Consider a example of an OTT platform for allocating resources, polling resources and monitoring the resources for the cloud services. With a neat labelled diagram Give the role of each agent associated.

Answer:

An **OTT platform** (like Netflix or Amazon Prime Video) delivers video content using cloud. To manage resources efficiently, it uses **agents** such as:

- Allocation Agent
- Polling Agent
- Monitoring Agent

Agents and Their Functions

1. Allocation Agent

- Allocates resources like servers, storage, and bandwidth
- Increases resources during high demand

Example:

When many users watch a new movie, more servers are added automatically

2. Polling Agent

- Checks system status regularly
- Collects data like CPU, memory, and server load

Example:

Checks every few seconds if streaming servers are working properly

3. Monitoring Agent

- Analyzes data from polling agent
- Sends alerts and triggers actions

Example:

If server load is high → automatically increases resources

Working of System

1. User requests video
2. Allocation agent provides resources
3. Polling agent checks system status
4. Monitoring agent analyzes and controls system

Real-Life Example:

In **Netflix** during a new release:

- Many users join at same time
- Allocation agent increases servers
- Polling agent checks performance
- Monitoring agent prevents overload

Result: Smooth streaming without buffering.

Agents help in:

- Efficient resource usage
- Continuous monitoring
- Better performance

Hence, agents are essential for cloud-based OTT platforms

18. What is resource replication. Is resource replication an ready-made environment a hand to hand approach for validating large resources in cloud environment. Justify with an example

Resource replication is the process of creating multiple instances (copies) of the same IT resource (such as virtual servers, storage, or applications) in a cloud environment to improve availability, performance, and reliability.

- It is usually implemented using virtualization technology.
- Virtual machine (VM) images can be easily copied, moved, or cloned.
- Replication can be done using methods like:
 - OVF (Open Virtualization Format)
 - Folder method (download & upload)
- Resource replication is not a ready-made environment.
- A ready-made environment refers to a pre-defined cloud platform (PaaS) that includes pre-installed tools, databases, and development kits (SDKs) ready for use. In contrast, resource replication is only a mechanism used to duplicate IT resources.
- However, it indirectly supports such environments by improving their reliability, scalability, and fault tolerance.

- Resource replication is also not a hand-to-hand (manual) approach for validating large resources.
- It is an automated cloud mechanism focused on resource management rather than validation.
- However, replicated resources can be used for testing and validation purposes.
- Large datasets or services can be tested on replicated copies without affecting the original system, which makes validation safer and more efficient, but validation itself is a separate process.

Example 1: Virtual Server Replication in Cloud

- A cloud provider stores a virtual server image
- The hypervisor creates multiple copies (replicas) of this server

Scenario:

- One VM → used for live application
- Another VM → used for backup
- Another VM → used for testing Benefits:
- If one server fails → others continue working
- Testing can be done on replicas without disturbing users
- System performance improves due to load distribution

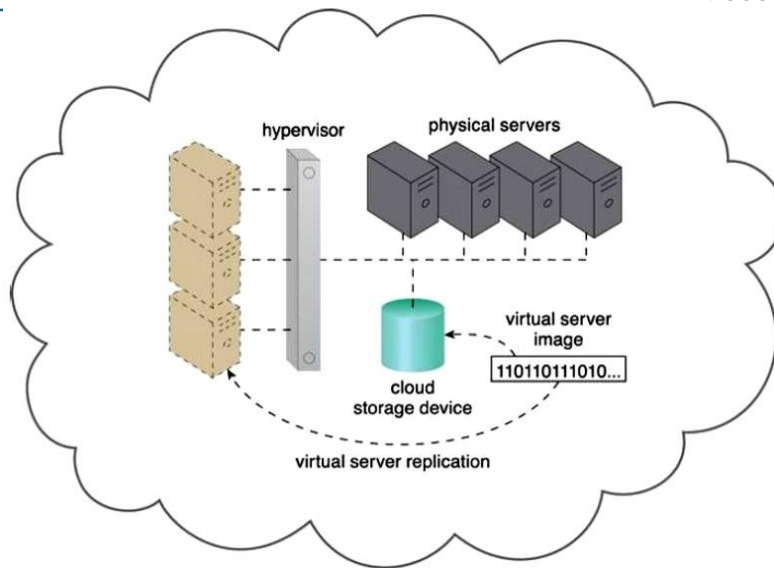
Example(real time): E-commerce Website (like Amazon-like system)

- Suppose a cloud server hosts a shopping website.
- During a sale (like Big Billion Day), traffic increases.
- Using resource replication:
- Multiple copies of the same server (VM) are created.
- Requests are distributed across these replicas.

Benefits:

- Faster response time
- No server crash
- High availability

If one server fails → others continue working (fault tolerance)



The hypervisor replicates several instances of a virtual server, using a stored virtual server image.